PRODUCT USER MANUAL

For Sea Surface Surface (SST) Products over the Mediterranean Sea

SST_MED_SST_L4_NRT_OBSERVATIONS_010_004
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
GHRSST	Group for High Resolution SST
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our Help Center is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document provides to Copernicus Marine users practical information on merged multi-sensor (level 3S, L3S) and optimally interpolated (level 4, L4) operational production of satellite-based Sea Surface Temperature (SST) data over the Mediterranean Sea (Table 1) processed by the Italian National Research Council-Institute of Marine Sciences (CNR-ISMAR).

Table 1. *Copernicus Marine Service - Mediterranean Sea Near Real Time L3S and L4 SST data.*

PRODUCT REFERENCE	FULL NAME	AREA	TYPE	LEVEL
SST_MED_SST_L4_NRT_ OBSERVATIONS_010_004	Mediterranean Sea - High Resolution and Ultra High-Resolution Sea Surface Temperature Analysis	MED	NRT	L4
SST_MED_SST_L3S_NRT_ OBSERVATIONS_010_012	Mediterranean Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature	MED	NRT	L3S

The Mediterranean Sea (MED) near-real-time (NRT) products include daily (nighttime) gridded fields of Sea Surface Temperature (SST), provided as both merged multi-sensor L3S and optimally interpolated L4 datasets, covering the entire Mediterranean basin. Both L3S and L4 products are available at high resolution (HR, 1/16°) and ultra-high resolution (UHR, 1/100°) and represent nighttime SST (centered at 00:00 UTC), thus providing the so-called foundation SST—that is, the temperature largely unaffected by diurnal warming (see e.g. Minnett et al., 2019).

The MED NRT SST products (Table 1) are produced daily selecting only the highest quality input L2 data within a short temporal window (local nighttime), to avoid cloud contamination and diurnal cycle.

To ensure the highest quality of the archived datasets, each product is reprocessed one month later, meaning that the SST field for a given day is regenerated 30 days after its initial production. This approach enables the use of a symmetric temporal window in the interpolation algorithm, improving both the consistency and accuracy of the final product.

The quality of these products (Table 1), along with a detailed description of the algorithm and the associated processing chain, is provided in the corresponding QUID (Pisano et al., 2025). Users are also referred to the published literature describing the SST MED algorithm (Buongiorno Nardelli et al., 2013) and its subsequent evolutions (Buongiorno Nardelli et al., 2015; Fanelli et al., 2024).

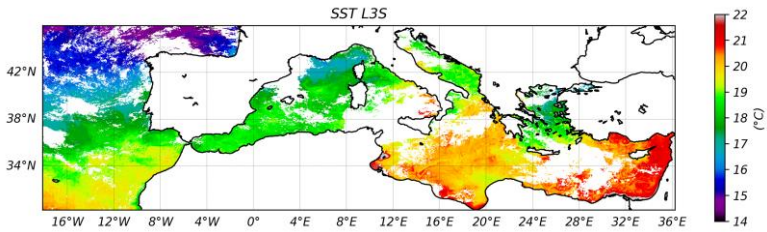
2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The MED NRT SST products (Table 1) are GHRSST-compliant, meaning their file format adheres to the GHRSST Data Specifications (GDS[1]). This format is widely adopted by the SST community, including major data providers such as space agencies.

The area encompassing the adjacent North Atlantic Ocean is included to support assimilation activities carried out by the MED Monitoring and Forecasting Centre (MFC). Both the HR and UHR SST products are accompanied by a daily SST anomaly dataset, representing deviations from a baseline climatology calculated over the 1985–2005 period. A synthesis of the MED NRT SST products characteristic is reported in Table 2 and Table 3, while a complete list and description of all datasets available within the MED NRT SST product family is provided in the dataset details section.

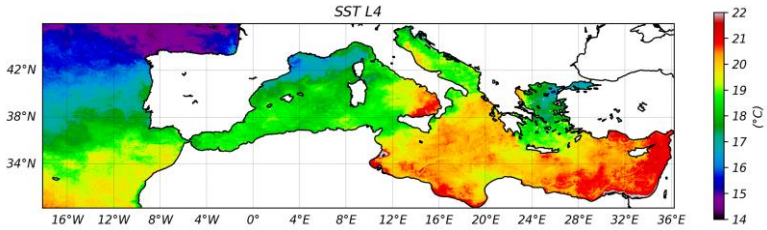
***Table 2**. Copernicus Marine Service - Mediterranean Sea NRT L3S SST product specifications.*

Product Lines	SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012
Geographical coverage	Mediterranean Sea + adjacent North Atlantic box  • HR - [W=-18.125, E=36.325, S=30.125, N=46.025] • UHR - [W=-18.121, E=36.246, S=30.254, N=45.996]
Variables	• Sea Surface Temperature • Adjusted Sea Surface Temperature
Product Type	Near Real Time – L3S
Available time series	01/01/2008 - present
Temporal resolution	Daily (nighttime)
Update Frequency	Daily
Target delivery time	Daily at 12:00 (noon) UTC for HR and 15:00 UTC for UHR
Horizontal resolution	• HR - 0.0625° x 0.0625° (1/16°) ; regular grid ; lon x lat = 871 x 253 • UHR - 0.01° x 0.01° (1/100°), regular grid, lon x lat = 6525 x 1890

[1] GHRSST - GHRSST Project Documents

Number of vertical levels	Surface product
Format	NetCDF4 – CF-1.4 convention compliant

Table 3. *Copernicus Marine Service – Mediterranean Sea NRT L4 SST product specifications.*

Product Lines	SST_MED_SST_L4_NRT_OBSERVATIONS_010_004
Geographical coverage	Mediterranean Sea + adjacent North Atlantic box  • HR – [W=-18.125, E=36.325, S=30.125, N=46.025] • UHR – [W=-18.121, E=36.246, S=30.254, N=45.996]
Variables	• Analysed Sea Surface Temperature • Analysis error
Product Type	Near Real Time – L4
Available time series	01/01/2008 – present
Temporal resolution	Daily (nighttime)
Update Frequency	Daily
Target delivery time	Daily at 12 :00 (noon) UTC for HR and 15 :00 UTC for UHR
Horizontal resolution	• HR – 0.0625° x 0.0625° (1/16°) ; regular grid ; lon x lat = 871 x 253 • UHR - 0.01° x 0.01° (1/100°), regular grid, lon x lat = 6525 x 1890
Number of vertical levels	Surface product
Format	NetCDF4 – CF-1.4 convention compliant

b) Production System Description

The MED NRT SST operational system runs daily at the CNR-ISMAR facility, starting at 07:00 UTC. Products are typically made available to users at 12:00 UTC for HR, at 15:00 UTC for UHR (see Table 2).

Both L3S and L4 products are derived from nighttime observations acquired by infrared sensors onboard multiple satellite platforms. The MED NRT SST processing chain involves several steps—from data extraction and initial quality control, to cloud screening and multi-sensor image merging. L3S fields are generated by selecting only the highest quality Level-2 data (quality flag 5) acquired during a narrow local nighttime window, from 21:00 of the day before to 06:00 of the interpolation day (local pixel time), effectively minimizing the impact of both cloud contamination and diurnal warming.

A two-step interpolation algorithm is then applied to generate L4 fields at both HR and UHR, using statistical techniques (Buongiorno Nardelli et al., 2013). A 10-day window (corresponding to 10 L3S fields) is used to generate the L4 HR product, while a 3-day window is applied for the UHR product.

The optimal interpolation scheme uses different first-guess fields and covariance models depending on the product, i.e. HR or UHR.

The MED HR L4 algorithm employs a daily pentad climatology (based on the 1985–2005 period) as its first-guess field, constructed from 21 years of AVHRR Pathfinder data (Marullo et al., 2007). The key elements of the algorithm—including the first-guess computation, covariance model, data sub-sampling strategy, and inversion technique—are detailed in Buongiorno Nardelli et al. (2013). Since April 2014, the algorithm has been upgraded with a new covariance function model, featuring a revised analytical formulation and spatially variable parameters. Further details are provided in the corresponding QUID (Pisano et al., 2025) while a full description of the algorithm's development and validation are presented in Buongiorno Nardelli et al. (2015). Additionally, from 2024 onwards, an improved first-guess field has been used in the generation of the MED UHR L4 data, enhancing the product's spatial resolution of SST features and the accuracy of SST gradients via machine learning techniques (Fanelli et al., 2024).

These L4 fields are also used to compute SST anomalies relative to a pentad climatology whose reference period is 1985-2005.

The L2P infrared observations currently used to generate the MED SST products are acquired from the following satellite sensors (Figure 1): the Sea and Land Surface Temperature Radiometer (SLSTR) on board Sentinel-3A and Sentinel-3B; the Visible Infrared Imaging Radiometer Suite (VIIRS) on board NOAA-21, NOAA-20, and Suomi National Polar-orbiting Partnership (Suomi NPP); the Advanced Very High Resolution Radiometer (AVHRR) on board Metop-B and Metop-C; the Moderate Resolution Imaging Spectroradiometer (MODIS) on board AQUA and TERRA; and the Spinning Enhanced Visible and InfraRed Imager (SEVIRI) on board the Meteosat Second Generation (MSG) satellite. Figure 1 illustrates the timeline of sensor integration into the MED SST processing chain from 2018 onwards.

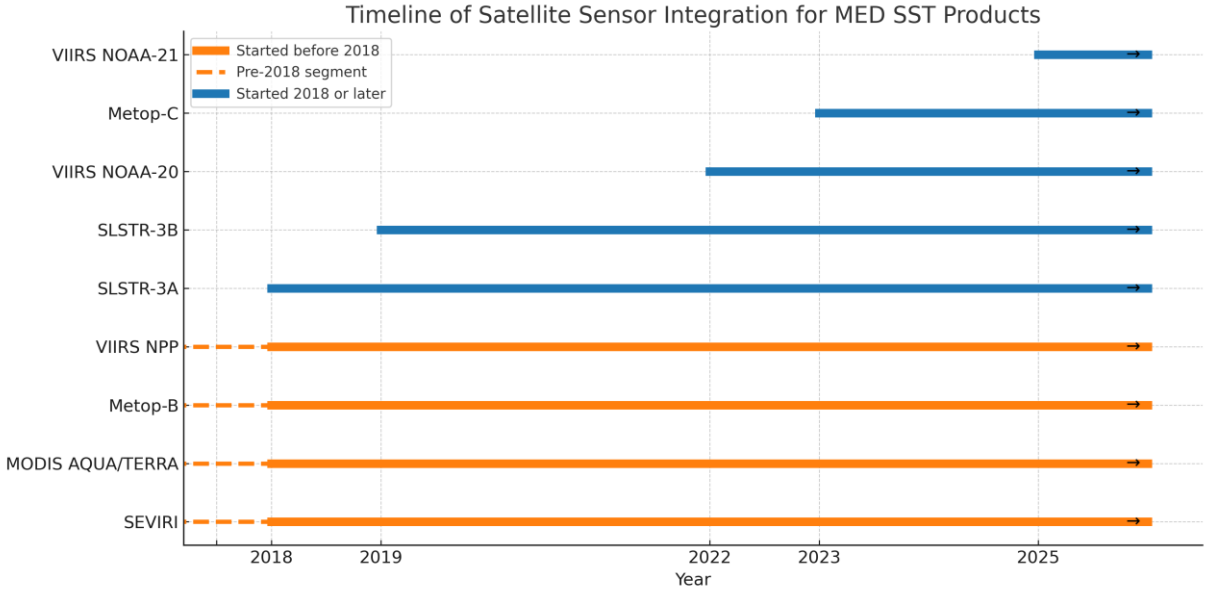


Figure 1. Timeline of sensor integration into the MED NRT SST processing chain from 2018 onwards.

c) Dataset details

Table 4. *List of the four datasets that make up the SST_MED_SST_L4_NRT_OBSERVATIONS_010_004 product. For each dataset, the main variable name and unist are provided, along with the long name, standard name, and a brief comment.*

SST_MED_SST_L4_NRT_OBSERVATIONS_010_004
Dataset: SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2. This dataset provides the daily (nighttime) SST analysis (L4), centered at 00:00 UTC, along with an estimate of SST uncertainty, over the Mediterranean Sea at high spatial resolution (HR = 0.0625°).
analysed_sst [K] analysed_sst:long_name = "analysed sea surface temperature" ; analysed_sst:standard_name = "sea_surface_foundation_temperature" ; Mediterranean SST analysis at high spatial resolution.
analysis_error [K] analysis_error:long_name = "estimated error standard deviation of analysed_sst" ; Estimated error standard deviation of analysed_sst.
mask mask:long_name = "sea/land/lake/ice field composite mask"; Land, sea-ice, lake bit mask
Dataset: SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2. This dataset provides the daily (nighttime) SST analysis (L4), centered at 00:00 UTC, along with an estimate of SST uncertainty, over the Mediterranean Sea at ultra high spatial resolution (HR = 0.01°).
analysed_sst [K] analysed_sst:long_name = "analysed sea surface temperature" ; analysed_sst:standard_name = "sea_surface_foundation_temperature" ; Mediterranean SST analysis at ultra high spatial resolution
analysis_error [K] analysis_error:long_name = "estimated error standard deviation of analysed_sst" ; Estimated error standard deviation of analysed_sst.

mask mask:long_name = "sea/land/lake/ice field composite mask"; Land, sea-ice, lake bit mask
Dataset: SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_b. This dataset provides the daily (nighttime) SST anomaly (L4) over the Mediterranean Sea at high spatial resolution (HR = 0.0625°). The anomalies are computed with respect to the 1985–2005 reference baseline.
sst_anomaly [K]; sst_anomaly:long_name = "sea surface temperature anomaly"; sst_anomaly:standard_name = "sea_surface_temperature_anomaly"; SST anomaly: computed as the difference between the L4 HR data and a corresponding pentad climatology.
Dataset: SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_d. This dataset provides the daily (nighttime) SST anomaly (L4) over the Mediterranean Sea at high spatial resolution (UHR = 0.01°). The anomalies are computed with respect to the 1985–2005 reference baseline.
sst_anomaly [K]; sst_anomaly:long_name = "sea surface temperature anomaly"; sst_anomaly:standard_name = "sea_surface_temperature_anomaly"; SST anomaly: computed as the difference between the L4 UHR data and a corresponding pentad climatology.

***Table 5**. List of the two datasets that make up the* SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012 *product. For each dataset, the main variable name and unit are provided, along with the long name, standard name, and a brief comment* List of the main variable names and unit for the SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012 product.

SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012
Dataset: SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a. This dataset provides the daily (nighttime) merged multi-sensor (L3S) SST field, centered at 00 :00 UTC, over the Mediterranean Sea at high (HR=0.0625°) spatial resolution.
adjusted_sea_surface_temperature [K] adjusted_sea_surface_temperature:long_name = "adjusted sea surface foundation_temperature"; adjusted_sea_surface_temperature:standard_name = "sea_surface_foundation_temperature"; Sea surface temperature corrected with a bias adjustment relative to a reference sensor. This is the recommended SST value for use.
sea_surface_temperature [K] sea_surface_temperature:long_name = "sea surface temperature" ; sea_surface_temperature:standard_name = "sea_surface_foundation_temperature" ; Uncorrected sea surface temperature values, i.e. taken directly from the input source products and merged into this dataset, without applying any bias adjustment.
sst_dtime [second] sst_dtime:long_name = "time difference from reference time"; Time difference of the observation value from the file reference time (centred at 00:00 UTC)
quality_level quality_level:long_name = "quality level of SST pixel"; Quality level of the SST value (copied from the source input). These are the overall quality indicators and are used for all GHRSST SSTs
sses_bias [K] sses_bias:long_name = "SSES bias error"; Bias estimate of the uncorrected sea surface temperature value, as provided in the source input.
sses_standard_deviation [K] sses_standard_deviation:long_name = "SSES standard deviation";

Error of the uncorrected sea surface temperature value, as provided in the source input.
or_latitude [degree north] or_latitude:long_name = "original latitude of the SST value";
or_longitude [degree east] or_longitude:long_name = "original longitude of the SST value";
or_number_of_pixels or_number_of_pixels:long_name = "number of pixels from the L2P contributing to the SST value";
sources_of_sst sources_of_sst:long_name = "SST product origin";
bias_to_reference_sst bias_to_reference_sst:long_name = "bias error derived from reference"; Bias error estimated to correct the original SST value.
Dataset: SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b. This dataset provides the daily (nighttime) merged multi-sensor (L3S) SST field, centered at 00:00 UTC, over the Mediterranean Sea at ultra high (HR=0.01°) spatial resolution.
adjusted_sea_surface_temperature [K] adjusted_sea_surface_temperature:long_name = "adjusted sea surface foundation_temperature"; adjusted_sea_surface_temperature:standard_name = "sea_surface_foundation_temperature"; Sea surface temperature corrected with a bias adjustment relative to a reference sensor. This is the recommended SST value for use.
sea_surface_temperature [K] sea_surface_temperature:long_name = "sea surface temperature" ; sea_surface_temperature:standard_name = "sea_surface_foundation_temperature" ; Uncorrected sea surface temperature values, i.e. taken directly from the input source products and merged into this dataset, without applying any bias adjustment.
sst_dtime [second] sst_dtime:long_name = "time difference from reference time"; Time difference of the observation value from the file reference time (centred at 00:00 UTC)
quality_level quality_level:long_name = "quality level of SST pixel"; Quality level of the SST value (copied from the source input). These are the overall quality indicators and are used for all GHRSST SSTs
sses_bias [K] sses_bias:long_name = "SSES bias error"; Bias estimate of the uncorrected sea surface temperature value, as provided in the source input.
sses_standard_deviation [K] sses_standard_deviation:long_name = "SSES standard deviation"; Error of the uncorrected sea surface temperature value, as provided in the source input.
or_latitude [degree north] or_latitude:long_name = "original latitude of the SST value";
or_longitude [degree east] or_longitude:long_name = "original longitude of the SST value";
or_number_of_pixels or_number_of_pixels:long_name = "number of pixels from the L2P contributing to the SST value";
sources_of_sst sources_of_sst:long_name = "SST product origin";
bias_to_reference_sst bias_to_reference_sst:long_name = "bias error derived from reference"; Bias error estimated to correct the original SST value.

d) Additional information on some parameters

SST_MED_SST_L4_NRT_OBSERVATIONS_010_004

- Real_Value = (Display_Value X scale_factor) + add_offset
- The missing value for this product is: -32768s (analysed_sst and analysis_error)
- Land locations are set to the missing value, which is stored as the "_FillValue" variable attribute in the files.
- Inland water bodies, e.g. small lakes, are masked out.

SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012

- Real_Value = (Display_Value X scale_factor) + add_offset
- The missing value for this product is: -32768s (adjusted_sea_surface_temperature and sea_surface_temperature)
- Land locations are set to the missing value, which is stored as the "_FillValue" variable attribute in the files.
- Inland water bodies, e.g. small lakes, are masked out.

e) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Disseminat ion day
Analysis	N.A.	Daily (nighttime)	12:00 UTC for MED HR; 15:00 UTC for MED UHR	Every day

f) Catalogue updating

The MED SST products (Table 1) are updated daily by 12:00 (noon) UTC for HR and 15:00 UTC for UHR datasets.

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily	Each daily SST field is representative of nighttime SST values collected between 21:00 of the previous day and 06:00 of the interpolation day and centered at 00:00 UTC.	N/A

h) Product use and impact

For applications such as daily SST monitoring and near real-time analysis, there are several compelling reasons to prefer the products specifically developed for the Mediterranean Sea over global SST products. The main advantages include:

- Use of state-of-the-art algorithms tailored to the Mediterranean region, resulting in higher accuracy compared to global products over the same area.
- Enhanced spatial resolution, allowing for improved representation of oceanographic features such as temperature gradients, fronts, upwelling areas and nearshore dynamics.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

Dataset Name	File Name	Dimension [KB]
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	<date>000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_NRT-MED-v02.0-fv03.0.nc	250
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_C_V2	<date>000000-GOS-L4_GHRSST-SSTfnd-OISST_UHR_NRT-MED-v02.0-fv03.0.nc	5600
SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_b	<date> 000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_NRT_ANOM-MED-v02.0-fv02.nc	160
SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_d	<date> 000000-GOS-L4_GHRSST-SSTfnd-OISST_UHR_NRT_ANOM-MED-v02.0-fv02.nc	4400
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	<date>20240101000000-GOS-L3S_GHRSST-SSTsubskin-night_SST_HR_NRT-MED-v02.0-fv02.0.nc	990
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	<date>000000-GOS-L3S_GHRSST-SSTsubskin-night_SST_UHR_NRT-MED-v02.0-fv02.0.nc	26000

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Buongiorno Nardelli, B., Pisano, A., Tronconi, C., & Santoleri, R. (2015). Evaluation of different covariance models for the operational interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean Sea. Remote Sensing of Environment, 164, 334-343. https://doi.org/10.1016/j.rse.2015.04.025

Buongiorno Nardelli, B., Tronconi, C., Pisano, A., & Santoleri, R. (2013). High and Ultra-High resolution processing of satellite Sea Surface Temperature data over Southern European Seas in the framework of MyOcean project. Remote Sensing of Environment, 129, 1-16. https://doi.org/10.1016/j.rse.2012.10.012

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Minnett, P. J., Alvera-Azcárate, A., Chin, T. M., Corlett, G. K., Gentemann, C. L., Karagali, I., ... & Vazquez-Cuervo, J. (2019). Half a century of satellite remote sensing of sea-surface temperature. Remote Sensing of Environment, 233, 111366. https://doi.org/10.1016/j.rse.2019.111366

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6) ANNEX

For dataset SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2

```
netcdf  20240101000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_NRT-MED-v02.0-fv03.0
{
dimensions:
        time = 1 ;
        lat = 253 ;
        lon = 871 ;
variables:
       int time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:calendar = "Gregorian" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:comment = "Nominal time of Level 4 analysis" ;
        float lon(lon) ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
                lon:long_name = "longitude" ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
                lon:valid_min = -20.f ;
                lon:valid_max = 40.f ;
        float lat(lat) ;
                lat:comment = "geographical coordinates, WGS84 projection" ;
                lat:long_name = "latitude" ;
                lat:standard_name = "latitude" ;
                lat:axis = "Y" ;
                lat:units = "degrees_north" ;
                lat:valid_min = 29.f ;
                lat:valid_max = 47.f ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:long_name = "analysed sea surface temperature" ;
                analysed_sst:standard_name = "sea_surface_foundation_temperature" ;
                analysed_sst:type = "foundation" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:add_offset = 273.15f ;
                analysed_sst:scale_factor = 0.01f ;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
                analysed_sst:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-ATS_NR_2P,JPL-L2P-
MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-NAR18_SST,EUR-L2P-
SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-AVHRR_METOP_B,NAVO-L2P-
VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-SLSTR_S3A,MAR-L2P-
SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-AVHRR_METOP_C,STAR-
L2P-VIIRS_N21-ACSPO_V2.80" ;
                analysed_sst:comment = "Optimal interpolation (OI) at high resolution of sst
measurements from infrared sensors" ;
                analysed_sst:_FillValue = -32768s ;
        short analysis_error(time, lat, lon) ;
```

```
            analysis_error:long_name = "estimated error standard deviation of analysed_sst" ;
            analysis_error:units = "kelvin" ;
            analysis_error:add_offset = 0.f ;
            analysis_error:scale_factor = 0.01f ;
            analysis_error:valid_min = 0s ;
            analysis_error:valid_max = 10000s ;
            analysis_error:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-ATS_NR_2P,JPL-L2P-
MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-NAR18_SST,EUR-L2P-
SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-AVHRR_METOP_B,NAVO-L2P-
VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-SLSTR_S3A,MAR-L2P-
SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-AVHRR_METOP_C,STAR-
L2P-VIIRS_N21-ACSPO_V2.80" ;
            analysis_error:comment = "Error obtained by Optimal interpolation (OI) at high
resolution of sst measurements from infrared sensors" ;
            analysis_error:_FillValue = -32768s ;
        byte sea_ice_fraction(time, lat, lon) ;
            sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
            sea_ice_fraction:long_name = "sea ice area fraction" ;
            sea_ice_fraction:units = "1" ;
            sea_ice_fraction:add_offset = 0.f ;
            sea_ice_fraction:scale_factor = 0.01f ;
            sea_ice_fraction:valid_min = 0b ;
            sea_ice_fraction:valid_max = 100b ;
            sea_ice_fraction:source_ice = "Not applicabile for Mediterranean sea" ;

            sea_ice_fraction:comment_ice = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:comment = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:source = "Not applicabile for Mediterranean sea" ;
            sea_ice_fraction:_FillValue = -128b ;
        byte mask(time, lat, lon) ;
            mask:long_name = "sea/land/lake/ice field composite mask" ;
            mask:units = "1" ;
            mask:flag_values = 1b, 2b, 3b, 4b ;
            mask:flag_meanings = "sea land lake ice" ;
            mask:valid_min = 1b ;
            mask:valid_max = 4b ;
            mask:source = "---" ;
            mask:comment = "b0:1= sea; b1:1= land; b2:1= lake; b3:1= sea ice; b4-7: reserved
for grid mask" ;
            mask:_FillValue = -128b ;

// global attributes:
            :Conventions = "CF-1.4" ;
            :title = "Mediterranean SST Analysis, L4, 1/16deg daily
(SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2)" ;
            :summary = "Daily gap-free maps (L4) at 0.0625deg. x 0.0625deg. horizontal
resolution over the Mediterranean Sea. The data (for day=dd) are obtained from infra-red
measurements collected [since 7pm of day=dd-1 to 7am of day=dd] by satellite radiometers
and statistical interpolation.It is the Copernicus sea surface temperature nominal operational
product for the Mediterranean Sea in NRT mode." ;
            :references = "Buongiorno Nardelli, B., C. Tronconi, a. Pisano, and R. Santoleri,
2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data
over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./,
*129*, 1-16, doi:10.1016/j.rse.2012.10.012. AND Buongiorno Nardelli, B., a. Pisano, C.
```

Tronconi, and R. Santoleri, 2015: Evaluation of different covariance models for the operational interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025." ;
:institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem study)" ;
:history = "GOS-MYO processor V3: new version" ;
:comment = "WARNING: some applications are unable to properly handle byte values. If Values >127 are encounterd, please subtract 256" ;
:license = "free registration at Copernicus Marine Service (http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
:id = "OISST_HR_NRT-GOS-L4-MED-v3.0" ;
:naming_authority = "org.ghrsst" ;
:product_version = "1.1" ;
:uuid = "f74e8ae0-a888-11ee-9c0b-005056be6f27" ;
:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20240101T093431Z" ;
:file_quality_level = 0 ;
:spatial_resolution = "0.0625 degree" ;
:start_time = "20231231T190000Z" ;
:time_coverage_start = "20231231T190000Z" ;
:stop_time = "20240101T070000Z" ;
:time_coverage_end = "20240101T070000Z" ;
:software_version = "Copernicus Marine Service HR L4 Processor V.1.0" ;
:northernmost_latitude = 46.f ;
:southernmost_latitude = 30.25f ;
:easternmost_longitude = 36.25f ;
:westernmost_longitude = -18.125f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;
:source = "EUR-L2P-ATS_NR_2P, UPA-L2P-ATS_NR_2P, JPL-L2P-MODIS_A, JPL-L2P-MODIS_T, EUR-L2P-NAR17_SST, EUR-L2P-NAR18_SST, EUR-L2P-SEVIRI_SST, EUR-L2P-AVHRR_METOP_A, EUR-L2P-AVHRR_METOP_B, NAVO-L2P-VIIRS_NPP, EUR-L2P-SEVIRI_SST_Meteosat11, MAR-L2P-SLSTR_S3A, MAR-L2P-SLSTR_S3B, STAR-L2P-VIIRS_N20-ACSPO_V2.80, EUR-L2P-AVHRR_METOP_C, STAR-L2P-VIIRS_N21-ACSPO_V2.80" ;
:platform = "Envisat, NOAA-18, MetOpA, Aqua, Terra, MSG1, MetOpB, SUOMI-NPP, Meteosat_11, Sentinel 3A, Sentinel 3B, NOAA-20, MetOpC, NOAA-21" ;
:sensor = "AATSR, SEVIRI, AVHRR, MODIS, VIIRS, SLSTR" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lat_resolution = 0.0625f ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lon_resolution = 0.0625f ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We would also appreciate being informed of any publications." ;

```
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
}
```

For dataset SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2

```
netcdf  20240101000000-GOS-L4_GHRSST-SSTfnd-OISST_UHR_NRT-MED-v02.0-fv03.0
{
dimensions:
      time = 1 ;
      lat = 1890 ;
      lon = 6525 ;
variables:
      int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "Gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "Nominal time of Level 4 analysis" ;
      float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
            lon:units = "degrees_east" ;
            lon:comment = "geographical coordinates, WGS84 projection" ;
            lon:valid_min = -20.f ;
            lon:valid_max = 40.f ;
      float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
            lat:comment = "geographical coordinates, WGS84 projection" ;

            lat:valid_min = 29.f ;
            lat:valid_max = 47.f ;
      short analysed_sst(time, lat, lon) ;
            analysed_sst:long_name = "analysed sea surface temperature" ;
            analysed_sst:standard_name = "sea_surface_foundation_temperature" ;
            analysed_sst:type = "foundation" ;
            analysed_sst:units = "kelvin" ;
            analysed_sst:add_offset = 273.15f ;
            analysed_sst:scale_factor = 0.01f ;
            analysed_sst:valid_min = -300s ;
            analysed_sst:valid_max = 4500s ;
```

```
            analysed_sst:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-ATS_NR_2P,JPL-L2P-
MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-NAR18_SST,EUR-L2P-
SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-AVHRR_METOP_B,NAVO-L2P-
VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-SLSTR_S3A,MAR-L2P-
SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-AVHRR_METOP_C ,STAR-
L2P-VIIRS_N21-ACSPO_V2.80" ;
            analysed_sst:comment = "Optimal interpolation (OI) at Ultrahigh resolution of sst
measurements from infrared sensors and an AI-based super resolved first guess" ;
            analysed_sst:_FillValue = -32768s ;
        short analysis_error(time, lat, lon) ;
            analysis_error:long_name = "estimated error standard deviation of analysed_sst" ;
            analysis_error:units = "kelvin" ;
            analysis_error:add_offset = 0.f ;
            analysis_error:scale_factor = 0.01f ;
            analysis_error:valid_min = 0s ;
            analysis_error:valid_max = 10000s ;
            analysis_error:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-ATS_NR_2P,JPL-L2P-
MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-NAR18_SST,EUR-L2P-
SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-AVHRR_METOP_B,NAVO-L2P-
VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-SLSTR_S3A,MAR-L2P-
SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-AVHRR_METOP_C ,STAR-
L2P-VIIRS_N21-ACSPO_V2.80" ;
            analysis_error:comment = "Error obtained by Optimal interpolation (OI) at Ultrahigh
resolution of sst measurements from infrared sensors" ;
            analysis_error:_FillValue = -32768s ;
        byte sea_ice_fraction(time, lat, lon) ;
            sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
            sea_ice_fraction:long_name = "sea ice area fraction" ;
            sea_ice_fraction:units = "1" ;
            sea_ice_fraction:add_offset = 0.f ;

            sea_ice_fraction:scale_factor = 0.01f ;
            sea_ice_fraction:valid_min = 0b ;
            sea_ice_fraction:valid_max = 100b ;
            sea_ice_fraction:source_ice = "Not applicabile for Mediterranean Sea" ;
            sea_ice_fraction:comment_ice = "Not applicabile for Mediterranean Sea" ;
            sea_ice_fraction:source = "Not applicabile for Mediterranean Sea" ;
            sea_ice_fraction:comment = "Not applicabile for Mediterranean Sea" ;
            sea_ice_fraction:_FillValue = -128b ;
        byte mask(time, lat, lon) ;
            mask:long_name = "sea/land/lake/ice field composite mask" ;
            mask:units = "1" ;
            mask:flag_values = 1b, 2b, 3b, 4b ;
            mask:flag_meanings = "sea land lake ice" ;
            mask:valid_min = 1b ;
            mask:valid_max = 4b ;
            mask:source = "----" ;
            mask:comment = "b0:1= sea; b1:1= land; b2:1= lake; b3:1= sea ice; b4-7: reserved
for grid mask" ;
            mask:_FillValue = -128b ;

    // global attributes:
            :Conventions = "CF-1.4" ;
```

:title = "Mediterranean SST Analysis, L4, 1km daily (SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2)" ;
:summary = "Daily gap-free maps (L4) at 1/100deg. x 1/100deg. horizontal resolution over the Mediterranean Sea. The data (for day=dd) are obtained from infra-red measurements collected [since 7pm of day=dd-1 to 7am of day=dd] by satellite radiometers and statistical interpolation.It is the Copernicus sea surface temperature nominal operational product for the Mediterranean Sea in DT mode." ;
:references = "Buongiorno Nardelli, B., Tronconi, C., Pisano A., and Santoleri R., 2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./, *129*, 1-16, doi:10.1016/j.rse.2012.10.012. AND Buongiorno Nardelli, B., Pisano A., Tronconi C., and Santoleri R., 2015: Evaluation of different covariance models for the operational interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025. AND Fanelli, C., Ciani, D., Pisano, A., and Buongiorno Nardelli, B., 2024: Deep learning for the super resolution of Mediterranean sea surface temperature fields. /Ocean Science/, 20(4), 1035-1050, doi:10.5194/os-20-1035-2024." ;
:institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem study)" ;
:history = "GOS-MYO processor V3: new version" ;
:comment = "WARNING: some applications are unable to properly handle byte values. If Values >127 are encounterd, please subtract 256" ;
:license = "free registration at Copernicus Marine Service (http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
:id = "OISST_UHR_NRT-GOS-L4-MED-v3.0" ;
:naming_authority = "org.ghrsst" ;
:product_version = "1.1" ;
:uuid = "706b2c92-ae75-11ef-a9f6-005056be6f27" ;
:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20241129T171449Z" ;
:file_quality_level = 0 ;
:spatial_resolution = "0.01 degree" ;
:start_time = "20231231T190000Z" ;
:time_coverage_start = "20231231T190000Z" ;
:stop_time = "20240101T070000Z" ;
:time_coverage_end = "20240101T070000Z" ;
:software_version = "Copernicus Marine Service UHR L4 Processor V.2" ;
:westernmost_longitude = -18.12083f ;
:easternmost_longitude = 36.24583f ;
:southernmost_latitude = 30.25417f ;
:northernmost_latitude = 45.99583f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;
:source = "EUR-L2P-ATS_NR_2P, UPA-L2P-ATS_NR_2P, JPL-L2P-MODIS_A, JPL-L2P-MODIS_T, EUR-L2P-NAR17_SST, EUR-L2P-NAR18_SST, EUR-L2P-SEVIRI_SST, EUR-L2P-AVHRR_METOP_A, EUR-L2P-AVHRR_METOP_B, NAVO-L2P-VIIRS_NPP, EUR-L2P-SEVIRI_SST_Meteosat11, MAR-L2P-SLSTR_S3A ,MAR-L2P-SLSTR_S3B, STAR-L2P-VIIRS_N20-ACSPO_V2.80, EUR-L2P-AVHRR_METOP_C , STAR-L2P-VIIRS_N21-ACSPO_V2.80" ;
:platform = "Envisat, NOAA-18, MetOpA, Aqua, Terra, MSG1, MetOpB, SUOMI-NPP, Meteosat_11, Sentinel 3A, Sentinel 3B, NOAA-20, MetOpC , NOAA-21" ;
:sensor = "AATSR, SEVIRI, AVHRR, MODIS, VIIRS, SLSTR" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;

```
            :keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
            :keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science
Keywords" ;
            :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata
Convention" ;
            :geospatial_lat_units = "degrees_north" ;

            :geospatial_lat_resolution = 0.01f ;
            :geospatial_lon_units = "degrees_east" ;
            :geospatial_lon_resolution = 0.01f ;
            :acknowledgment = "Please acknowledge the use of these data with the following
statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We
would also appreciate being informed of any publications." ;
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
}
```

For dataset SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_b
netcdf 20240101000000-GOS-L4_GHRSST-SSTfnd-OISST_HR_NRT_ANOM-MED-v02.0-fv02.0.nc

```
{
dimensions:
        time = 1 ;
        lat = 253 ;
        lon = 871 ;
variables:
        int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "Gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "Nominal time of Level 4 analysis" ;
        float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
            lon:units = "degrees_east" ;
            lon:comment = "geographical coordinates, WGS84 projection" ;

        float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
```

```
            lat:comment = "geographical coordinates, WGS84 projection" ;
        short sst_anomaly(time, lat, lon) ;
            sst_anomaly:long_name = "sea surface temperature anomaly" ;
            sst_anomaly:standard_name = "sea_surface_temperature_anomaly" ;
            sst_anomaly:type = "" ;
            sst_anomaly:units = "Celsius" ;
            sst_anomaly:comment = "SST anomaly: difference between L4 HR Optimal
interpolated data and a pentad climatology" ;
            sst_anomaly:add_offset = 0.f ;
            sst_anomaly:scale_factor = 0.01f ;
            sst_anomaly:valid_min = -300s ;
            sst_anomaly:valid_max = 4500s ;
            sst_anomaly:_FillValue = -32768s ;

// global attributes:
            :Conventions = "CF-1.4" ;
            :title = "Mediterranean SST Anomaly, L4, 1/16deg daily
(SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_b)" ;
            :summary = "SST anomaly: difference between L4 HR Optimal interpolated data
and a pentad climatology" ;
            :DSD_entry_id = "-GOS-L4HRfnd-MED" ;
            :references = "Buongiorno Nardelli, B., C. Tronconi, a. Pisano, and R. Santoleri,
2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data
over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./,
*129*, 1-16, doi:10.1016/j.rse.2012.10.012. AND Buongiorno Nardelli, B., a. Pisano, C.
Tronconi, and R. Santoleri, 2015: Evaluation of different covariance models for the operational
interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean
Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025." ;
            :institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem
study)" ;
            :comment = "WARNING: some applications are unable to properly handle byte
value. If Value are encounterd > 127, please subtract 256 from this reported value" ;
            :license = "free registration at Copernicus Marine Service
(http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
            :id = "OISST_HR_NRT_ANOM-GOS-L4-MED-v2.0" ;
            :naming_authority = "org.ghrsst" ;
            :product_version = "2.0" ;
            :uuid = "0ea54066-3664-11f0-9f65-005056be6f27" ;
            :GDS_version_id = "v2.0-rev 5" ;
            :netcdf_version_id = "4.7.1, Build Date: OCT  6 2022" ;
            :date_created = "20250521T165301Z" ;
            :file_quality_index = 0 ;
            :grid_resolution = " 0.0625 degree" ;
            :software_version = "Copernicus Marine Service L4 Processor V.1.1" ;
            :acknowledgment = "Please acknowledge the use of these data with the following
statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We
would also appreciate being informed of any publications." ;
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
```

```
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu, gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
            :distribution_statement = "See Copernicus Marine Service Data License" ;
            :westermost_longitude = -18.125f ;
            :eastermost_longitude = 36.25f ;
            :southermost_latitude = 30.25f ;
            :northermost_latitude = 46.f ;
            :start_date = "2023-12-31" ;
            :start_time = "21:00:00" ;
            :stop_date = "2024-01-01" ;
            :stop_time = "06:00:00" ;
}
```

For dataset SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_d
netcdf 20240101000000-GOS-L4_GHRSST-SSTfnd-OISST_UHR_NRT_ANOM-MED-v02.0-fv02.0.nc

```
{
dimensions:
        time = 1 ;
        lat = 1890 ;
        lon = 6525 ;

variables:
        int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "Gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "Nominal time of Level 4 analysis" ;
        float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
            lon:units = "degrees_east" ;
            lon:comment = "geographical coordinates, WGS84 projection" ;
        float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
            lat:comment = "geographical coordinates, WGS84 projection" ;
        short sst_anomaly(time, lat, lon) ;
            sst_anomaly:long_name = "sea surface temperature anomaly" ;
            sst_anomaly:standard_name = "sea_surface_temperature_anomaly" ;
            sst_anomaly:type = "" ;
            sst_anomaly:units = "Celsius" ;
            sst_anomaly:comment = "SST anomaly: difference between L4 UHR Optimal interpolated data using an AI-based super resolved first guess and a pentad climatology" ;
            sst_anomaly:add_offset = 0.f ;
            sst_anomaly:scale_factor = 0.01f ;
```

```
            sst_anomaly:valid_min = -300s ;
            sst_anomaly:valid_max = 4500s ;
            sst_anomaly:_FillValue = -32768s ;

// global attributes:
            :Conventions = "CF-1.4" ;
            :title = "Mediterranean SST Anomaly, L4, 1km daily
(SST_MED_SSTA_L4_NRT_OBSERVATIONS_010_004_d)" ;
            :summary = "SST anomaly: difference between L4 UHR Optimal interpolated data
using an AI-based super resolved first guess and a pentad climatology" ;
            :DSD_entry_id = "-GOS-L4UHRfnd-MED" ;

            :references = "Buongiorno Nardelli, B., Tronconi, C., Pisano A., and Santoleri R.,
2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data
over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./,
*129*, 1-16, doi:10.1016/j.rse.2012.10.012. AND Buongiorno Nardelli, B., Pisano A., Tronconi
C., and Santoleri R., 2015: Evaluation of different covariance models for the operational
interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean
Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025. AND Fanelli, C., Ciani, D.,
Pisano, A., and Buongiorno Nardelli, B., 2024: Deep learning for the super resolution of
Mediterranean sea surface temperature fields. /Ocean Science/, 20(4), 1035-1050,
doi:10.5194/os-20-1035-2024." ;
            :institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem
study)" ;
            :comment = "WARNING: some applications are unable to properly handle byte
value. If Value are encounterd > 127, please subtract 256 from this reported value" ;
            :license = "free registration at Copernicus Marine Service
(http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
            :id = "OISST_UHR_NRT_ANOM-GOS-L4-MED-v2.0" ;
            :naming_authority = "org.ghrsst" ;
            :product_version = "2.0" ;
            :uuid = "820bbbd4-3664-11f0-91f8-005056be6f27" ;
            :GDS_version_id = "v2.0-rev 5" ;
            :netcdf_version_id = "4.7.1, Build Date: OCT  6 2022" ;
            :date_created = "20250521T165615Z" ;
            :file_quality_index = 0 ;
            :grid_resolution = "0.01 degree" ;
            :software_version = "Copernicus Marine Service L4 Processor V.2.0" ;
            :acknowledgment = "Please acknowledge the use of these data with the following
statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We
would also appreciate being informed of any publications." ;
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L4" ;
            :cdm_data_type = "grid" ;
            :distribution_statement = "See Copernicus Marine Service Data License" ;
            :westermost_longitude = -18.12083f ;
            :eastermost_longitude = 36.24583f ;
```

```
            :southermost_latitude = 30.25417f ;
            :northermost_latitude = 45.99583f ;
            :start_date = "2023-12-31" ;
            :start_time = "21:00:00" ;
            :stop_date = "2024-01-01" ;
            :stop_time = "06:00:00" ;
}
```

For dataset SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a
netcdf 20240101000000-GOS-L3S_GHRSST-SSTsubskin-night_SST_HR_NRT-MED-v02.0-fv02.0.nc

```
{
dimensions:
        time = 1 ;
        lat = 253 ;
        lon = 871 ;
variables:
        int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "Gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "Nominal time of Level 4 analysis" ;
        float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
            lon:units = "degrees_east" ;
            lon:comment = "geographical coordinates, WGS84 projection" ;
            lon:valid_min = -20.f ;
            lon:valid_max = 40.f ;
            lon:reference_datum = "geographical coordinates, equirectangular projection" ;
        float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
            lat:comment = "geographical coordinates, WGS84 projection" ;
            lat:valid_min = 29.f ;

            lat:valid_max = 47.f ;
            lat:reference_datum = "geographical coordinates, equirectangular projection" ;
        int sst_dtime(time, lat, lon) ;
            sst_dtime:long_name = "time difference from reference time" ;
            sst_dtime:units = "seconds" ;
            sst_dtime:comment = "Variable time plus sst_dtime gives seconds after 00:00:00
UTC January 1, 1981" ;
            sst_dtime:add_offset = 0 ;
            sst_dtime:scale_factor = 1 ;
            sst_dtime:valid_min = -32767 ;
            sst_dtime:valid_max = 32767 ;
            sst_dtime:_FillValue = -32768 ;
        short sea_surface_temperature(time, lat, lon) ;
```

```
            sea_surface_temperature:long_name = "sea surface temperature" ;
            sea_surface_temperature:standard_name = "sea_surface_foundation_temperature"
;
            sea_surface_temperature:units = "kelvin" ;
            sea_surface_temperature:add_offset = 273.15f ;
            sea_surface_temperature:scale_factor = 0.01f ;
            sea_surface_temperature:valid_min = -300s ;
            sea_surface_temperature:valid_max = 4500s ;
            sea_surface_temperature:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-
ATS_NR_2P,JPL-L2P-MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-
NAR18_SST,EUR-L2P-SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-
AVHRR_METOP_B,NAVO-L2P-VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-
SLSTR_S3A,MAR-L2P-SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-
AVHRR_METOP_C ,STAR-L2P-VIIRS_N21-ACSPO_V2.80" ;
            sea_surface_temperature:comment = "SST measurement values from input L2
satellite data set." ;
            sea_surface_temperature:_FillValue = -32768s ;
      byte quality_level(time, lat, lon) ;
            quality_level:long_name = "quality level of SST pixel" ;
            quality_level:flag_meanings = "no_data bad_data worst_quality low_quality
acceptable_quality best_quality" ;
            quality_level:flag_values = 0b, 1b, 2b, 3b, 4b, 5b ;
            quality_level:units = "1" ;
            quality_level:valid_min = 0b ;
            quality_level:valid_max = 5b ;
            quality_level:source = "----" ;
            quality_level:comment = "These are the overall quality indicators and are used for all
GHRSST SSTs" ;

            quality_level:_FillValue = -128b ;
      byte sses_bias(time, lat, lon) ;
            sses_bias:long_name = "SSES bias error" ;
            sses_bias:units = "kelvin" ;
            sses_bias:add_offset = 0.f ;
            sses_bias:scale_factor = 0.01f ;
            sses_bias:valid_min = -127b ;
            sses_bias:valid_max = 127b ;
            sses_bias:source = "---" ;
            sses_bias:comment = "---" ;
            sses_bias:_FillValue = -128b ;
      byte sses_standard_deviation(time, lat, lon) ;
            sses_standard_deviation:long_name = "SSES standard deviation" ;
            sses_standard_deviation:units = "kelvin" ;
            sses_standard_deviation:add_offset = 1.f ;
            sses_standard_deviation:scale_factor = 0.01f ;
            sses_standard_deviation:valid_min = -127b ;
            sses_standard_deviation:valid_max = 127b ;
            sses_standard_deviation:source = "---" ;
            sses_standard_deviation:comment = "---" ;
            sses_standard_deviation:_FillValue = -128b ;
      short adjusted_sea_surface_temperature(time, lat, lon) ;
            adjusted_sea_surface_temperature:long_name = "adjusted sea surface
foundation_temperature" ;
```

```
            adjusted_sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
            adjusted_sea_surface_temperature:units = "kelvin" ;
            adjusted_sea_surface_temperature:add_offset = 273.15f ;
            adjusted_sea_surface_temperature:scale_factor = 0.01f ;
            adjusted_sea_surface_temperature:valid_min = -300s ;
            adjusted_sea_surface_temperature:valid_max = 4500s ;
            adjusted_sea_surface_temperature:reference = "MAR-L2P-SLSTR_S3A" ;
            adjusted_sea_surface_temperature:comment = "priorities : example : NAVO-L2P-
VIIRS_NPP, ATS_NR_2P, AVHRRMTA, NAR17_SST, NAR18_SST, AVHRR17_L,
AVHRR_18_L, AVHRR17_L, AVHRR18_G, SEVIRI_1H_SST, GOES_12_1H_SST, AMSRE,
TMI, MODIS_A, MODIS_T, METOP_A, METOP_B, METOP_C" ;
            adjusted_sea_surface_temperature:source = "---" ;
            adjusted_sea_surface_temperature:_FillValue = -32768s ;
        short or_latitude(time, lat, lon) ;
            or_latitude:long_name = "original latitude of the SST value" ;
            or_latitude:standard_name = "latitude" ;

            or_latitude:units = "degrees_north" ;
            or_latitude:add_offset = 0.f ;
            or_latitude:scale_factor = 0.01f ;
            or_latitude:valid_min = 2900s ;
            or_latitude:valid_max = 4700s ;
            or_latitude:source = "---" ;
            or_latitude:comment = "Original latitude of the SST value" ;
            or_latitude:_FillValue = -32768s ;
        short or_longitude(time, lat, lon) ;
            or_longitude:long_name = "original longitude of the SST value" ;
            or_longitude:standard_name = "longitude" ;
            or_longitude:units = "degree_east" ;
            or_longitude:add_offset = 0.f ;
            or_longitude:scale_factor = 0.01f ;
            or_longitude:valid_min = -2000s ;
            or_longitude:valid_max = 4000s ;
            or_longitude:source = "---" ;
            or_longitude:comment = "Original longitude of the SST value" ;
            or_longitude:_FillValue = -32768s ;
        short or_number_of_pixels(time, lat, lon) ;
            or_number_of_pixels:long_name = "number of pixels from the L2Ps contributing to
the SST value" ;
            or_number_of_pixels:units = "1" ;
            or_number_of_pixels:add_offset = 0.f ;
            or_number_of_pixels:scale_factor = 1.f ;
            or_number_of_pixels:valid_min = 0s ;
            or_number_of_pixels:valid_max = 32767s ;
            or_number_of_pixels:source = "---" ;
            or_number_of_pixels:comment = "---" ;
            or_number_of_pixels:_FillValue = -32768s ;
        float sum_sst(time, lat, lon) ;
            sum_sst:long_name = "sum of contributing pixel sst values" ;
            sum_sst:units = "kelvin" ;
            sum_sst:add_offset = 0.f ;
            sum_sst:scale_factor = 1.f ;
            sum_sst:valid_min = 0.f ;
```

```
            sum_sst:valid_max = 1.e+10f ;
            sum_sst:source = "---" ;
            sum_sst:comment = "Sum of orginal contributing pixel sst values" ;
            sum_sst:_FillValue = -99.f ;


        float sum_square_sst(time, lat, lon) ;
            sum_square_sst:long_name = "sum of contributing pixel sst value squares" ;
            sum_square_sst:units = "kelvin^2" ;
            sum_square_sst:add_offset = 0.f ;
            sum_square_sst:scale_factor = 1.f ;
            sum_square_sst:valid_min = 0.f ;
            sum_square_sst:valid_max = 1.e+20f ;
            sum_square_sst:source = "---" ;
            sum_square_sst:comment = "Sum of contributing pixel sst value squares" ;
            sum_square_sst:_FillValue = -99.f ;
        byte sources_of_sst(time, lat, lon) ;
            sources_of_sst:long_name = "SST product origin" ;
            sources_of_sst:units = "1" ;
            sources_of_sst:add_offset = 0.f ;
            sources_of_sst:scale_factor = 1.f ;
            sources_of_sst:valid_min = -127b ;
            sources_of_sst:valid_max = 127b ;
            sources_of_sst:comment = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3,
METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9,
MODIS_T :10, SEVIRI :11, " ;
            sources_of_sst:source = "---" ;
            sources_of_sst:_FillValue = -128b ;
        byte adjusted_standard_deviation_error(time, lat, lon) ;
            adjusted_standard_deviation_error:long_name = "standard deviation error based on
SSES and adjustment method" ;
            adjusted_standard_deviation_error:units = "kelvin" ;
            adjusted_standard_deviation_error:add_offset = 0.f ;
            adjusted_standard_deviation_error:scale_factor = 0.01f ;
            adjusted_standard_deviation_error:valid_min = -127b ;
            adjusted_standard_deviation_error:valid_max = 127b ;
            adjusted_standard_deviation_error:comment = "Cumulated errors of SSES and
adjustment method" ;
            adjusted_standard_deviation_error:source = "---" ;
            adjusted_standard_deviation_error:_FillValue = -128b ;
        short bias_to_reference_sst(time, lat, lon) ;
            bias_to_reference_sst:long_name = "bias error derived from reference" ;
            bias_to_reference_sst:units = "kelvin" ;
            bias_to_reference_sst:add_offset = 0.f ;
            bias_to_reference_sst:scale_factor = 0.01f ;
            bias_to_reference_sst:valid_min = -32767s ;
            bias_to_reference_sst:valid_max = 32767s ;
            bias_to_reference_sst:comment = "Bias estimate derived from comparison between
the original SST (native SSES being applied) and the reference sensor SST (original SST -
reference SST)" ;
            bias_to_reference_sst:source = "---" ;
            bias_to_reference_sst:_FillValue = -32768s ;
        byte standard_deviation_to_reference_sst(time, lat, lon) ;
```

```
            standard_deviation_to_reference_sst:long_name = "standard_deviation error
derived from reference" ;
            standard_deviation_to_reference_sst:units = "kelvin" ;
            standard_deviation_to_reference_sst:add_offset = 0.f ;
            standard_deviation_to_reference_sst:scale_factor = 1.f ;
            standard_deviation_to_reference_sst:valid_min = -127b ;
            standard_deviation_to_reference_sst:valid_max = 127b ;
            standard_deviation_to_reference_sst:comment = "This represents the error
standard deviation estimate resulting from the bias estimation method" ;
            standard_deviation_to_reference_sst:source = "---" ;
            standard_deviation_to_reference_sst:_FillValue = -128b ;
        byte wind_speed(time, lat, lon) ;
            wind_speed:long_name = "wind_speed" ;
            wind_speed:standard_name = "wind_speed" ;
            wind_speed:units = "m s-1" ;
            wind_speed:height = "10 m" ;
            wind_speed:add_offset = 0.f ;
            wind_speed:scale_factor = 1.f ;
            wind_speed:valid_min = -127b ;
            wind_speed:valid_max = 127b ;
            wind_speed:time_offset = 0.f ;
            wind_speed:coordinates = "---" ;
            wind_speed:comment = "These wind speeds were created by the ECMWF and
represent winds at 10 metres above the sea surface" ;
            wind_speed:source = "ECMWF_Analysis_V2" ;
            wind_speed:_FillValue = -128b ;
        short l2p_flags(time, lat, lon) ;
            l2p_flags:long_name = "L2P flags" ;
            l2p_flags:units = "1" ;
            l2p_flags:flag_meanings = "microwave land ice lake river sun_glint
SST_algorithm_A SST_algorithm_B SST_algorithm_C SST_algorithm_D" ;
            l2p_flags:flag_masks = 1s, 2s, 4s, 8s, 16s, 32s, 64s, 128s, 256s, 512s ;
            l2p_flags:add_offset = 0.f ;
            l2p_flags:scale_factor = 1.f ;
            l2p_flags:valid_min = 0s ;

            l2p_flags:valid_max = 32767s ;
            l2p_flags:time_offset = 0s ;
            l2p_flags:coordinates = "lon lat" ;
            l2p_flags:comment = "These flags are important to properly use the data. The tables
below define the bit field and their meanings: Bit 0, Flags=0 if infrared data,1 if passive
microwave data - Bit 1, Flags=0 if over the ocean, 1 if over land - Bit 2  Ice: Set if pixel is over
ice - Bit 3  Lake: Set if pixel is over a lake (if known) - Bit 4  River: Set if pixel is over a river (if
known) - Bit 5  Reserved for future use - Bit 6-15: Defined by L2 data provider." ;
            l2p_flags:source = "---" ;
            l2p_flags:_FillValue = -32768s ;

// global attributes:
            :Conventions = "CF-1.4" ;
            :title = "Mediterranean SST, L3S, 1/16deg daily
(SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a)" ;
            :summary = "The MED CNR Sea Surface Temperature L3S products provide daily
super-collated maps of SST at 0.0625deg. x 0.0625deg. horizontal resolution over the
```

Mediterranean Sea.The data are obtained from infra-red measurements collected by satellite radiometers and specific super-collating procedures in NRT mode" ;
:references = "Buongiorno Nardelli, B., C. Tronconi, a. Pisano, and R. Santoleri, 2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./, *129*, 1-16, doi:10.1016/j.rse.2012.10.012. and Buongiorno Nardelli, B., a. Pisano, C. Tronconi, and R. Santoleri, 2015: Evaluation of different covariance models for the operational interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025." ;
:institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem study)" ;
:history = "GOS-MYO processor V3: new version" ;
:comment = "WARNING: the last 5 variable are still in progress - WARNING: some applications are unable to properly handle byte values. If Values >127 are encounterd, please subtract 256" ;
:license = "free registration at Copernicus Marine Service (http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
:id = "OISST_HR_NRT-GOS-L3S-MED-v2.0" ;
:DSD_entry_id = "-GOS-L3S_GHRSST-SSTsubskin" ;
:naming_authority = "org.ghrsst" ;
:product_version = "1.1" ;
:uuid = "f774374a-a888-11ee-b376-005056be6f27" ;
:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20240101T093431Z" ;
:file_quality_level = 0 ;
:spatial_resolution = "0.0625 degree" ;
:start_time = "20231231T190000Z" ;
:time_coverage_start = "20231231T190000Z" ;
:stop_time = "20240101T070000Z" ;
:time_coverage_end = "20240101T070000Z" ;
:software_version = "Copernicus Marine Service HR L3 Processor V.1.0" ;
:northernmost_latitude = 46.f ;
:southernmost_latitude = 30.25f ;
:easternmost_longitude = 36.25f ;
:westernmost_longitude = -18.125f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;
:source_data = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3, METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9, MODIS_T :10, SEVIRI :11, " ;
:source = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3, METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9, MODIS_T :10, SEVIRI :11, " ;
:platform = "Envisat, NOAA-18, MetOpA, Aqua, Terra, MSG1, MetOpB, SUOMI-NPP, Meteosat11, Sentinel 3A, Sentinel 3B, NOAA-20, MetOpC , NOAA-21" ;
:sensor = "AATSR, AVHRR, AVHRR_GAC, AVHRR_HRPT, SEVIRI, MODIS, VIIRS, SLSTR" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention" ;

```
            :geospatial_lat_units = "degrees_north" ;
            :geospatial_lat_resolution = 0.0625f ;
            :geospatial_lon_units = "degrees_east" ;
            :geospatial_lon_resolution = 0.0625f ;
            :acknowledgment = "Please acknowledge the use of these data with the following
statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We
would also appreciate being informed of any publications." ;
            :creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
            :creator_email = "gsdk@artov.ismar.cnr.it" ;
            :creator_url = "http://gosweb.artov.isac.cnr.it/" ;
            :project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;

            :publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
            :publisher_url = "http://marine.copernicus.eu/" ;
            :publisher_email = "servicedesk.cmems@mercator-ocean.eu,
gsdk@artov.ismar.cnr.it" ;
            :processing_level = "L3" ;
            :cdm_data_type = "grid" ;
            :Citation = "---" ;
            :distribution_statement = "See Copernicus Marine Service Data License" ;
}
```

For dataset SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b

netcdf 20240101000000-GOS-L3S_GHRSST-SSTsubskin-night_SST_UHR_NRT-MED-v02.0-fv02.0.nc

```
{
dimensions:
      time = 1 ;
      lat = 1890 ;
      lon = 6525 ;
variables:
      int time(time) ;
            time:long_name = "reference time of sst field" ;
            time:standard_name = "time" ;
            time:axis = "T" ;
            time:calendar = "Gregorian" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:comment = "Nominal time of Level 4 analysis" ;
      float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
            lon:units = "degrees_east" ;
            lon:comment = "geographical coordinates, WGS84 projection" ;
            lon:valid_min = -20.f ;
            lon:valid_max = 40.f ;
            lon:reference_datum = "geographical coordinates, equirectangular projection" ;
      float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
            lat:units = "degrees_north" ;
            lat:comment = "geographical coordinates, WGS84 projection" ;
```

```
            lat:valid_min = 29.f ;
            lat:valid_max = 47.f ;
            lat:reference_datum = "geographical coordinates, equirectangular projection" ;
      int sst_dtime(time, lat, lon) ;
            sst_dtime:long_name = "time difference from reference time" ;
            sst_dtime:comment = "Variable time plus sst_dtime gives seconds after 00:00:00
UTC January 1, 1981" ;
            sst_dtime:units = "seconds" ;
            sst_dtime:add_offset = 0 ;
            sst_dtime:scale_factor = 1 ;
            sst_dtime:valid_min = -32767 ;
            sst_dtime:valid_max = 32767 ;
            sst_dtime:_FillValue = -32768 ;
      short sea_surface_temperature(time, lat, lon) ;
            sea_surface_temperature:long_name = "sea surface temperature" ;
            sea_surface_temperature:standard_name = "sea_surface_foundation_temperature"
;
            sea_surface_temperature:units = "kelvin" ;
            sea_surface_temperature:add_offset = 273.15f ;
            sea_surface_temperature:scale_factor = 0.01f ;
            sea_surface_temperature:valid_min = -300s ;
            sea_surface_temperature:valid_max = 4500s ;
            sea_surface_temperature:source = "EUR-L2P-ATS_NR_2P,UPA-L2P-
ATS_NR_2P,JPL-L2P-MODIS_A,JPL-L2P-MODIS_T,EUR-L2P-NAR17_SST,EUR-L2P-
NAR18_SST,EUR-L2P-SEVIRI_SST,EUR-L2P-AVHRR_METOP_A,EUR-L2P-
AVHRR_METOP_B,NAVO-L2P-VIIRS_NPP,EUR-L2P-SEVIRI_SST_Meteosat11,MAR-L2P-
SLSTR_S3A,MAR-L2P-SLSTR_S3B,STAR-L2P-VIIRS_N20-ACSPO_V2.80,EUR-L2P-
AVHRR_METOP_C ,STAR-L2P-VIIRS_N21-ACSPO_V2.80" ;
            sea_surface_temperature:comment = "SST measurement values from input L2
satellite data set." ;
            sea_surface_temperature:_FillValue = -32768s ;
      byte quality_level(time, lat, lon) ;
            quality_level:long_name = "quality level of SST pixel" ;
            quality_level:flag_meanings = "no_data bad_data worst_quality low_quality
acceptable_quality best_quality" ;
            quality_level:flag_values = 0b, 1b, 2b, 3b, 4b, 5b ;
            quality_level:units = "1" ;
            quality_level:valid_min = 0b ;
            quality_level:valid_max = 5b ;
            quality_level:source = "---" ;

            quality_level:comment = "These are the overall quality indicators and are used for all
GHRSST SSTs" ;
            quality_level:_FillValue = -128b ;
      byte sses_bias(time, lat, lon) ;
            sses_bias:long_name = "SSES bias error" ;
            sses_bias:units = "kelvin" ;
            sses_bias:add_offset = 0.f ;
            sses_bias:scale_factor = 0.01f ;
            sses_bias:valid_min = -127b ;
            sses_bias:valid_max = 127b ;
            sses_bias:source = "---" ;
            sses_bias:comment = "---" ;
            sses_bias:_FillValue = -128b ;
```

```
        byte sses_standard_deviation(time, lat, lon) ;
                sses_standard_deviation:long_name = "SSES standard deviation" ;
                sses_standard_deviation:units = "kelvin" ;
                sses_standard_deviation:add_offset = 1.f ;
                sses_standard_deviation:scale_factor = 0.01f ;
                sses_standard_deviation:valid_min = -127b ;
                sses_standard_deviation:valid_max = 127b ;
                sses_standard_deviation:source = "---" ;
                sses_standard_deviation:comment = "---" ;
                sses_standard_deviation:_FillValue = -128b ;
        short adjusted_sea_surface_temperature(time, lat, lon) ;
                adjusted_sea_surface_temperature:long_name = "adjusted sea surface foundation
temperature" ;
                adjusted_sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
                adjusted_sea_surface_temperature:units = "kelvin" ;
                adjusted_sea_surface_temperature:add_offset = 273.15f ;
                adjusted_sea_surface_temperature:scale_factor = 0.01f ;
                adjusted_sea_surface_temperature:valid_min = -300s ;
                adjusted_sea_surface_temperature:valid_max = 4500s ;
                adjusted_sea_surface_temperature:reference = "MAR-L2P-SLSTR_S3A" ;
                adjusted_sea_surface_temperature:comment = "priorities : example : NAVO-L2P-
VIIRS_NPP, ATS_NR_2P, AVHRRMTA, NAR17_SST, NAR18_SST, AVHRR17_L,
AVHRR_18_L, AVHRR17_L, AVHRR18_G, SEVIRI_1H_SST, GOES_12_1H_SST, AMSRE,
TMI, MODIS_A, MODIS_T, METOP_A, METOP_B, METOP_C" ;
                adjusted_sea_surface_temperature:source = "---" ;
                adjusted_sea_surface_temperature:_FillValue = -32768s ;


        short or_latitude(time, lat, lon) ;
                or_latitude:long_name = "original latitude of the SST value" ;
                or_latitude:standard_name = "latitude" ;
                or_latitude:units = "degrees_north" ;
                or_latitude:add_offset = 0.f ;
                or_latitude:scale_factor = 0.01f ;
                or_latitude:valid_min = 2900s ;
                or_latitude:valid_max = 4700s ;
                or_latitude:source = "---" ;
                or_latitude:comment = "Original latitude of the SST value" ;
                or_latitude:_FillValue = -32768s ;
        short or_longitude(time, lat, lon) ;
                or_longitude:long_name = "original longitude of the SST value" ;
                or_longitude:standard_name = "longitude" ;
                or_longitude:units = "degree_east" ;
                or_longitude:add_offset = 0.f ;
                or_longitude:scale_factor = 0.01f ;
                or_longitude:valid_min = -2000s ;
                or_longitude:valid_max = 4000s ;
                or_longitude:source = "---" ;
                or_longitude:comment = "Original longitude of the SST value" ;
                or_longitude:_FillValue = -32768s ;
        short or_number_of_pixels(time, lat, lon) ;
                or_number_of_pixels:long_name = "number of pixels from the L2Ps contributing to
the SST value" ;
```

```
            or_number_of_pixels:units = "1" ;
            or_number_of_pixels:add_offset = 0.f ;
            or_number_of_pixels:scale_factor = 1.f ;
            or_number_of_pixels:valid_min = 0s ;
            or_number_of_pixels:valid_max = 32767s ;
            or_number_of_pixels:source = "---" ;
            or_number_of_pixels:comment = "---" ;
            or_number_of_pixels:_FillValue = -32768s ;
        float sum_sst(time, lat, lon) ;
            sum_sst:long_name = "sum of contributing pixel sst values" ;
            sum_sst:units = "kelvin" ;
            sum_sst:add_offset = 0.f ;
            sum_sst:scale_factor = 1.f ;
            sum_sst:valid_min = 0.f ;
            sum_sst:valid_max = 1.e+10f ;
            sum_sst:source = "---" ;
            sum_sst:comment = "Sum of orginal contributing pixel sst values" ;
            sum_sst:_FillValue = -99.f ;
        float sum_square_sst(time, lat, lon) ;
            sum_square_sst:long_name = "sum of contributing pixel sst value squares" ;
            sum_square_sst:units = "kelvin^2" ;
            sum_square_sst:add_offset = 0.f ;
            sum_square_sst:scale_factor = 1.f ;
            sum_square_sst:valid_min = 0.f ;
            sum_square_sst:valid_max = 1.e+20f ;
            sum_square_sst:source = "---" ;
            sum_square_sst:comment = "Sum of contributing pixel sst value squares" ;
            sum_square_sst:_FillValue = -99.f ;
        byte sources_of_sst(time, lat, lon) ;
            sources_of_sst:long_name = "SST product origin" ;
            sources_of_sst:units = "1" ;
            sources_of_sst:add_offset = 0.f ;
            sources_of_sst:scale_factor = 1.f ;
            sources_of_sst:valid_min = -127b ;
            sources_of_sst:valid_max = 127b ;
            sources_of_sst:comment = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3,
METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9,
MODIS_T :10, SEVIRI :11, " ;
            sources_of_sst:source = "---" ;
            sources_of_sst:_FillValue = -128b ;
        byte adjusted_standard_deviation_error(time, lat, lon) ;
            adjusted_standard_deviation_error:long_name = "standard deviation error based on
SSES and adjustment method" ;
            adjusted_standard_deviation_error:units = "kelvin" ;
            adjusted_standard_deviation_error:add_offset = 0.f ;
            adjusted_standard_deviation_error:scale_factor = 0.01f ;
            adjusted_standard_deviation_error:valid_min = -127b ;
            adjusted_standard_deviation_error:valid_max = 127b ;
            adjusted_standard_deviation_error:comment = "Cumulated errors of SSES and
adjustment method" ;
            adjusted_standard_deviation_error:source = "---" ;
            adjusted_standard_deviation_error:_FillValue = -128b ;
        short bias_to_reference_sst(time, lat, lon) ;
            bias_to_reference_sst:long_name = "bias error derived from reference" ;
```

```
            bias_to_reference_sst:units = "kelvin" ;
            bias_to_reference_sst:add_offset = 0.f ;
            bias_to_reference_sst:scale_factor = 0.01f ;
            bias_to_reference_sst:valid_min = -32767s ;

            bias_to_reference_sst:valid_max = 32767s ;
            bias_to_reference_sst:comment = "Bias estimate derived from comparison between
the original SST (native SSES being applied) and the reference sensor SST (original SST -
reference SST)" ;
            bias_to_reference_sst:source = "---" ;
            bias_to_reference_sst:_FillValue = -32768s ;
      byte standard_deviation_to_reference_sst(time, lat, lon) ;
            standard_deviation_to_reference_sst:long_name = "standard_deviation error
derived from reference" ;
            standard_deviation_to_reference_sst:units = "kelvin" ;
            standard_deviation_to_reference_sst:add_offset = 0.f ;
            standard_deviation_to_reference_sst:scale_factor = 1.f ;
            standard_deviation_to_reference_sst:valid_min = -127b ;
            standard_deviation_to_reference_sst:valid_max = 127b ;
            standard_deviation_to_reference_sst:comment = "This represents the error
standard deviation estimate resulting from the bias estimation method" ;
            standard_deviation_to_reference_sst:source = "---" ;
            standard_deviation_to_reference_sst:_FillValue = -128b ;
      byte wind_speed(time, lat, lon) ;
            wind_speed:long_name = "wind_speed" ;
            wind_speed:standard_name = "wind_speed" ;
            wind_speed:units = "m s-1" ;
            wind_speed:height = "10 m" ;
            wind_speed:add_offset = 0.f ;
            wind_speed:scale_factor = 1.f ;
            wind_speed:valid_min = -127b ;
            wind_speed:valid_max = 127b ;
            wind_speed:time_offset = 0.f ;
            wind_speed:coordinates = "---" ;
            wind_speed:comment = "These wind speeds were created by the ECMWF and
represent winds at 10 metres above the sea surface" ;
            wind_speed:source = "ECMWF_Analysis_V2" ;
            wind_speed:_FillValue = -128b ;
      short l2p_flags(time, lat, lon) ;
            l2p_flags:long_name = "L2P flags" ;
            l2p_flags:units = "1" ;
            l2p_flags:flag_meanings = "microwave land ice lake river sun_glint
SST_algorithm_A SST_algorithm_B SST_algorithm_C SST_algorithm_D" ;
            l2p_flags:flag_masks = 1s, 2s, 4s, 8s, 16s, 32s, 64s, 128s, 256s, 512s ;
            l2p_flags:add_offset = 0.f ;
            l2p_flags:scale_factor = 1.f ;

            l2p_flags:valid_min = 0s ;
            l2p_flags:valid_max = 32767s ;
            l2p_flags:time_offset = 0s ;
            l2p_flags:coordinates = "lon lat" ;
            l2p_flags:comment = "These flags are important to properly use the data The tables
below define the bit field and their meanings: Bit 0, Flags: 0 if infrared data,1 if passive
microwave data - Bit 1, Flags:0 if over the ocean, 1 if over land - Bit 2  Ice: Set if pixel is over
```

ice - Bit 3 Lake: Set if pixel is over a lake (if known) - Bit 4 River: Set if pixel is over a river (if known) - Bit 5 Reserved for future use - Bit 6-15: Defined by L2 data provider." ;
l2p_flags:source = "---" ;
l2p_flags:_FillValue = -32768s ;

// global attributes:
:Conventions = "CF-1.4" ;
:title = "Mediterranean SST, L3S, 1km daily (SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b)" ;
:summary = "The MED CNR Sea Surface Temperature L3S products provide daily super-collated maps of SST at 0.01deg. x 0.01deg. horizontal resolution over the Mediterranean Sea.The data are obtained from infra-red measurements collected by satellite radiometers and specific super-collating procedures in DT mode." ;
:references = "Buongiorno Nardelli, B., C. Tronconi, a. Pisano, and R. Santoleri, 2013: High and Ultra-High resolution processing of satellite Sea Surface Temperature data over Southern European Seas in the framework of MyOcean project. /Remote Sens. Environ./, *129*, 1-16, doi:10.1016/j.rse.2012.10.012. and Buongiorno Nardelli, B., a. Pisano, C. Tronconi, and R. Santoleri, 2015: Evaluation of different covariance models for the operational interpolation of high resolution satellite Sea Surface Temperature data over the Mediterranean Sea. /Remote Sens. Environ./, doi:10.1016/j.rse.2015.04.025." ;
:institution = "GOS (Global Ocean Satellite monitoring and marine ecosystem study)" ;
:history = "GOS-MYO processor V3 at UHR res: new version" ;
:comment = "WARNING: the last 5 variable are still in progress - WARNING: some applications are unable to properly handle byte values. If Values >127 are encounterd, please subtract 256" ;
:license = "free registration at Copernicus Marine Service (http://marine.copernicus.eu/web/56-user-registration-form.php)" ;
:id = "OISST_UHR_NRT-GOS-L3S-MED-v2.0" ;
:DSD_entry_id = "-GOS-L3S_GHRSST-SSTsubskin" ;
:naming_authority = "org.ghrsst" ;
:product_version = "1.0" ;
:uuid = "77daa26e-ae75-11ef-8bc2-005056be6f27" ;

:gds_version_id = "v2.0.5" ;
:netcdf_version_id = "4.7.1, Build Date: OCT 6 2022" ;
:date_created = "20240101T171501Z" ;
:file_quality_level = 0 ;
:spatial_resolution = "0.01 degree" ;
:start_time = "20231231T190000Z" ;
:time_coverage_start = "20231231T190000Z" ;
:stop_time = "20240101T070000Z" ;
:time_coverage_end = "20240101T070000Z" ;
:software_version = "Copernicus Marine Service UHR L3 Processor V.1.0" ;
:northernmost_latitude = 45.99583f ;
:southernmost_latitude = 30.25417f ;
:easternmost_longitude = 36.24583f ;
:westernmost_longitude = -18.12083f ;
:Scaling_Equation = "(scale_factor*data) + add_offset" ;
:source_data = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3, METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9, MODIS_T :10, SEVIRI :11, " ;

:source = "code: SLSTRA_D3 :1, VIIRS_NPP :2, METOP_C :3, METOP_B :4, VIIRS_N20 :5, SLSTRA_N3 :6, SLSTRB_D3 :7, SLSTRB_N3 :8, MODIS_A :9, MODIS_T :10, SEVIRI :11, " ;
:platform = "Envisat, NOAA-18, MetOpA, Aqua, Terra, MSG1, MetOpB, SUOMI-NPP, Meteosat11, Sentinel 3A, Sentinel 3B, NOAA-20, MetOpC , NOAA-21" ;
:sensor = "AATSR, AVHRR, AVHRR_GAC, AVHRR_HRPT, SEVIRI, MODIS, VIIRS, SLSTR" ;
:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
:metadata_link = "Link to collection metadata record at archive" ;
:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lat_resolution = 0.01f ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lon_resolution = 0.01f ;
:acknowledgment = "Please acknowledge the use of these data with the following statement: Generated/provided by Copernicus Marine Service and CNR - ISMAR ROME. We would also appreciate being informed of any publications." ;
:creator_name = "ISMAR - Institute of Marine Sciences (CNR - Rome)" ;
:creator_email = "gsdk@artov.ismar.cnr.it" ;

:creator_url = "http://gosweb.artov.isac.cnr.it/" ;
:project = "Copernicus Marine Environment Monitoring Service (CMEMS)" ;
:publisher_name = "CNR ISMAR GOS - CMEMS SST-TAC" ;
:publisher_url = "http://marine.copernicus.eu/" ;
:publisher_email = "servicedesk.cmems@mercator-ocean.eu, gsdk@artov.ismar.cnr.it" ;
:processing_level = "L3" ;
:cdm_data_type = "grid" ;
:Citation = "---" ;
:distribution_statement = "See Copernicus Marine Service Data License" ;
}